STRATEGIC FIT AND THE ORGANIZATION-ENVIRONMENT DEBATE

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The notion that organizations are shaped in fundamental ways by the nature of the environments in which they are embedded is not new. Barnard [1930] was perhaps the first to describe an open systems model of the organization and the process of administration. The idea did not give rise to much empirical work on organizational properties, however, until the late 1960s when work by Thompson [1967] and by Lawrence and Lorsch [1967] suggested that simplified constructs of environmental characteristics might still be theoretically interesting and productive of new knowledge. In the twelve years since those works were first published, the organization-environment paradigm has come to occupy an important, though not universally accepted, position in organizational sociology.

During these same twelve years, the strategy-policy perspective has come to moved to center stage in the fields of management and business policy. Like the growth of organization-environment models, the strategy-policy framework has taken root because it appears almost self-evident at a sufficiently high level of generalization; both frameworks can degenerate into purely situational arguments if the level of abstraction is made too low. The frameworks are not, however, identical. They differ particularly in the phenomena regarded as worthy of explanation, in the way in which the environmental contingency is modeled, and in the role organizational

structure plays in the whole process of adaptation. The purpose of this note is to highlight these differences and to discuss their implications.

The Strategy Concept

Organizational sociologists tend to associate the concept of strategy with Child's [1972] use of the managerial discretion argument in his attack on environmental determinism. Aldrich and Pfeffer's [1976] treatment of the "strategic choice" school of thought, for example, is essentially an attempt to state the conditions under which some degree of managerial choice is actually possible. Their conclusion on the issue of whether organizations can affect their environments is worth noting:

> Child's second criticism of the environmental selection model -- that organizations have the power to strategically influence their environments -- is true chiefly for the largest organizations or those politically well connected. However, only slightly more than 3% of all business enterprises, as enumerated by the US Social Security Administration, have over 50 employees. It is unlikely that firms of under this size have much power to affect their environments, although this varies by local circumstances.

More recently, Aldrich [1979:154] has added this coda to his discussion of strategic choice:

> The issue we must address, however, is whether it is fruitful to build a sociology of organizations on exceptional organizations ... Dominant organizations set conditions of existence for the small and powerless, and must be incorporated in any analysis of organizational change. Typically, however, they are not themselves the objects of analysis.

Our objection to this position is two-fold (and is not necessarily a defense of Child). First, and obviously, a theory that is restricted to the analysis of the powerless may have validity and usefulness, but we must not be surprised that its central core is one of environmental determinism and/or selection. It does not require long essays to justify an "external control" framework if the powerless are by definition the subject of analysis.

Secondly, and more crucially, the idea that only the powerful can alter their environment, while tautological as presented, stems from a sort of "geographic" vision of an organization embedded in some sort of field and changing that outside field by bringing some kind of force to bear and even then only with difficulty. If, however, we envision the environment as experienced through the particular transactions that are engaged in, it is clear that an organization's environment can be altered without altering the environment itself. To change environments one only has to do different things. The change is not simply cognitive, or perceptual, but quite physical.

A <u>small</u> firm, furthermore, can change its operating environment in many ways and has, perhaps, more flexibility in doing so than a large enterprise. For example, a small print shop might specialize in rapid service, night and weekend service, or special jobs. Instead, it might have close ties to a few large organizations and capitalize on its inside knowlede by providing service that is tailored to particular users's needs. Depending on the type of specialization pattern it fell into, our hypothetical print shop would face very different environments. One can easily imagine that the

problems of recruiting, training, and rewarding workers in an "overnite" print shop would be different from one which specialized in precision color work. Even in the unlikely event that both used the same type of machinery, they would have different production scheduling problems, differing needs for quality and cost control, and, very probably, would have different emergent social systems.

Modern biologists have made the same observation with regard to the relationship between organisms and their environments. As Lewontin [1978:215] notes,

> Organisms do not experience environments passively; they create and define the environment in which they live. Trees remake the soil in which they grow by dropping leaves and putting down roots ... Finally, organisms themselves determine which external factors will be part of their niche by their own activities. By building a nest the phoebe makes the availability of dried grass an important part of its niche, at the same time making the nest itself a component of the niche.

While Child's argument that "strategic choice is the critical variable in a theory of organizations" has usually been seen as a "managerial power" position by organizational theorists, that is not the core of the issue. In the strategy-policy field the concept of strategy need not be tied to rational planning or even conscious decision making assumptions. Strategy is essentially a descriptive idea that includes an organization's choice of niche and its primary decision rules (policies) for coping with that niche. The issue, then, is not managerial power, but whether differential selection processes operate directly on organizational form or on strategy. The strategy-policy framework strongly suggests that policy change is the

primary adaptive response of most organizations, that policy mix is the primary construct subject to selection, and that the impact of environment on organization is not direct but mediated by strategy.

The region of disagreement with either the population ecology or the resource dependence schools of thought is not, therefore, the nature of the selection process, as Aldrich and Pfeffer argued [1972], but the object being selected in the first place.

The system of logic that leads to this view is most appropriate in the context of the business firm and might best be termed the neoclassical concept of strategy. First developed by Chandler [1962] and Andrews [1964], the strategy concept places central attention on the relationships among policy, resources, and the envrironment. It places organizational structure in a variety of secondary roles -- as the repository of skill based resources, as a constraint on behavior, and as the means of carrying out policy.

The original strategy concept was heavily normative with strong implications that "fit" between the strategy and the environment was essential for survival. As empirical research has been carried out, however, it has become clearer that "fit" is not really an operational concept. Under rivalry, the dominant observable pattern is one of hetrogeneity rather than "fit". Strategy tends to consist of a process of domain differentiation rather than adaptation. Put another way, the environment does not appear to be "continuous" in the mathematical sense -- rather, it seems to be riddled with niches and subject to continuing topological change. Because neighboring

niches may present strikingly different problems to the organization, the primary issues facing most business firms are not "adaptation" in the large, but niche specialization and niche choice in the face of change.

Rivalry, Asymmetry, and Heterogeneity

In a rivalrous environment the outcomes of conflict are determined by asymmetries or diffences. The point appears self-evident, but has implications that are less obvious. Among them is the notion that models of environments which cluster firms according to commonly held structural or behavioral traits may easily conceal the phenomena of interest. Viewed from the comfortable distance of the IRS Source Book on Corporation Income, all beer firms may look pretty much alike. The relative success of individual beer firms, however, will be largely determined by their differences, and not their similarities. Thus, any model that is intended to capture important selection phenomena must deal with either (1) the important differences among the rivalrous entities, or (2) the dimensions along which important differences occur. Note that the need to deal with the fine structure of the environment is independent of whether or not managements are aware of its nature or even its existence.

Early empirical policy strategy research consisted of in-depth case studies of rivals within an industry. The Learned, Christensen, Andrews, and Guth [1965] casebook presented some of these for classroom use. The basic insight drawn from these cases was the heterogeneity within industries. Contrary to the models of industry

put forward in economics, the case studies showed close competitors differing in important ways and continually engaged in seeking out new sources of differentiation. While adaptation to a changing environment was a crucial issue for the "industry" as a whole, the individual firm faced the much more problematic question of doing it better or more efficiently than its rivals.

More recently, four streams of research have begun to converge, providing the beginning of a reconstructed theory of how firms actually compete and how competitive structures evolve over time. The four are:

1. The strategic groups concept as developed by Hunt [1975], Newman [1973], Porter [1974], and Caves and Porter [1977].

2. The disaggregation approaches of Bass [1974], Hatten [1974], and Patton [1971].

3. The market-share revealed-economies models with major contributions by Gale [1972], the Boston Consulting Group [1968], Schoeffler et al. [1972], and Demsetz [1973].

4. The "patterns" and "life-cycles" approaches to strategy identification developed and researched by Chandler [1962], Mintzberg [1978], Rumelt [1974], Miles and Snow [1978], and Miller and Friesen [1978].

The traditional model of industrial organization is almost apallingly simple: it is a one-parameter model that relates concentration to oligopolistic behavior and oligopolistic behavior to supra-normal levels of industry profitability. Often attacked for being overly simple and unrealistic, it has also suffered from a lack of strong verification despite numerous empirical tests spread over a twenty-five year period. The strategic groups theories are revised versions of traditional industrial economics. Their central theme

remains the impact of market structure on conduct and preformance, but they allow a wealth of new varieties of market structure to come under legitimate consideration. Porter, for example, has shown how manufacturer retailer interdependencies are a function of product type, advertising intensity, and the nature of the retailer. His theorizing was backed up by set of structural equation tests that were dramatically more successful than has been the norm in industrial organization research.

Bass, at Purdue, has led an econometric attact on homogeneity assumptions, especially those appearing in cross sectional studies across industries. Squarely in this camp, Hatten and Patton's work on the beer industry has revealed the fine structure within a supposedly homogeneous industry. Their careful disaggregation of the structural equations governing the profitability and market share of brewers has shown that there are at least three important subgroups within the beer industry. More recently, Hatten and Schendel [1977] have taken the analysis to the firm level and shown that heterogeneity continues to be revealed.

The third stream of empirical research begins to abandon the industry construct entirely and seeks to describe and measure the effects of niching at the firm or even product level. Its primary theoretical compenents are the experience effect, enuciated by Alchian [1959] and elaborated by the Boston Consulting Group [1968], and its major empirical result is the strong association between market share and profitability. While some have taken this association to imply a causal link, it is more commonly interpreted as evidence of either

experience based or scale based economies. What becomes most interesting, from the point of view of environmental modeling, is the potential use of the interactions between the market share profitability relationship and other variables as descriptors of the competitive environment.

Finally, the patterns or life cycles models of strategy have begun to provide us with alternative typologies of firms that group on the basis of commonly faced strategic dilemmas rather than static attributes held in common. Miles and Snow have focused on patterns of initiation and reaction, my own research dealt with the logics underlying strategies of diversification, and Mintzberg has developed proceedures for discerning strategy as pattern within an on going stream of events.[1]

Strategy and Organization

While strategy-policy had its origins in a "firm in the environment" view, the picture that has begun to emerge from longitudinal studies and cross sectional data is that of evolving competitive constellations. In part this is simply due to the recognition that key components of any firm's environment are other firms and rival firms at that. Additionally, however, it rests on the realization that rivalry is decomposable into two sets of activities: simple rivalry and strategic rivalry. Simple rivalry consists of niche maintence -- doing those things that serve to defend one's

[1] For a more complete description of the major themes of research in strategy-policy see Schendel and Hofer [1979].

domain, given the basic social, technological, and economic environment. Strategic rivalry, by contrast, involves changes in the niche structure itself. These changes normally arise due new technologies, new tastes, or changes in other related niche structures.

What are the implications of this new view for organization theory? While the full ramifications cannot yet be known, some preliminary insights and cautions can be presented.

Adaptation. If rivalry is a two-level process, the concept of adaptation, or fitness, becomes less clear. An organization can be well fitted to a niche, but if that niche itself is shifting or being forced out of existence, it is not, in a larger sense, dealing effectively with its environment.

Abernathy and Townsend's [1975] work on technological innovation and rivalry provides an illustration of this. They studied the phases of competitive behavior that followed major technological innovations and found three main stages. In the first phase competition focused on product characteristics. Cost was not typically the main bone of contention as rapid learning by both customers and producers increases the overall value created without much attention to efficiency per se. However, once a "dominant design" appeared, like the DC-3, the Model-T, or the Timex disposable watch, innovative activity was refocused on production technology. With less uncertainty in future material needs, vertical integration was undertaken and niche defense activity increased. However, this

phase led to increased specialization and an increasing asset base committed to a particular technology and product. The position was defensible as long as product innovation remained moderate. But significant new technologies, usually introduced by outsiders, faced the now highly specialized firm with a threat its organization was ill-equiped to cope with. Step-by-step the interplay between the innovative firm and its rivals had led it into a niche from which escape had become either impossible or hugely expensive.

If this multi-phase multi-level view of the adaptation and selection process has validity, it poses problems for both the contingency theorists and the population ecology models of organization. In the case of contingency theory, the primary dilemmas are these:

1. There no longer exists any simple connection between "fitness" and performance. Being fit to a dying niche is not much help yet the fact of strategic change does not mean that the competitive pressures that force niche specialization have diminished.

2. The idea that rivalry is about "something" in a given era and/or niche suggests that one is going to have to know much more about an environment than its degree of uncertainty or complexity if correlation with organizational structure is sought.

3. The changing basis of rivalry also suggests that organizations may well be loosely-coupled to strategy in some eras and more strongly coupled in others. While this would not invalidate a carefully stated contingency theory, it makes empirical verification much more difficult.

The population ecology model, due chiefly to Cambell [197?] and Hannan and Freeman [1977], has a close connection to microeconomics, an isomorphisism cogently discussed by Alchain [1950].

In its broad form, as a way of describing all natural phenomena, the model is really a philosophy of explanation that underlies much of Western science; science itself can be usefully described as a set of processes for generating, selecting, and retaining concepts about the natural world. Apparently powerful in its generality, the model is actually so general that it is virtually impossible to describe any phenomena that cannot be (post hoc) explained as the outcome of natural processes of variety generation, selection, and retention.

In its narrower form, the model seeks to explain the observed ordered variety in organizational forms as the outcome of those interactions with the external world which control the flow or denial of resources. The strategy policy model, however, suggests that selection processes do not act directly on organizational form. Rather, organizations are relatively stable entities that change (or experience differential selection on) their operating policies with much greater ease than they alter their structures. The first response of most organizations to environmental pressure is to alter an operating policy, not change structure. Thus, the presence of strategy as a mediating process between the organization per se and the environment weakens the expected strength of any empirical association between environmental features and organizational characteristics.

What, then, of the prospects for applying the population ecology model directly to strategy? The outlook for a more than metaphorical use is not bright. The experience to date in biology is that the behavioral adaptations possible in complex animals vastly

complicate the description of the selection processes acting on them. Organizational learning and human planning create the same complications in the social and economic worlds.

Resource Dependence. Mindlin and Aldrich's [1975:382] statement that "the major axiom of the resource dependence perspective is that organizations must be studied in the context of the population of organizations with which they are competing and sharing scarce resources" espouses a position identical to that taken by most strategy-policy theorists. The resource dependence model, as described by Aldrich [1979] and by Pfeffer and Salancik [1978], is closer in spirit to the policy strategy view than most other branches of organization theory. In part this is because of the willingness of the resource dependence researchers to investigate organizational actions as well as structure and in part is follows from a common concern with the utilization of critical resources.

Despite similarities, however, the resource dependence perspective tends to gloss over the question of what constitutes a strategic resource or a strategic dependency. The changing nature of what constitutes an advantage is, by contrast, one of the central themes in strategy-policy. Consider, for example, Pfeffer's [1972] hypothesis that firms tend to use merger to gain control over strategic resources (or to reduce critical dependencies). Strategy theorists might easily put forward the same hypothesis; the difference in viewpoint is revealed by the test employed. Pfeffer's test associated inter-industry merger rates with inter-industry transaction rates (input output coefficients). He found that firms do indeed tend

to merge with either their customers or their suppliers. From a strategy perspective the result was obvious; random merger activity would be remarkable, but certainly not mergers between two links in a vertical production chain. Merger is an event and dependence is a relationship; from a strategy perspective, the thing that needs explaining is why the merger occured at that time and in that industry. What shifts made those particular sets of relationships into "strategic dependencies?"

Conclusions

Serious research in strategy-policy began at about the same time that the organization envrionment perspective came into currency. Strategy-policy has tended to produce more detailed accounts of environments and more elaborate models of collective rivalry while the organization-environment field has made structural attributes the center of attention and typically employed rather simple models of the environment.

Organizational theorists tend to associate the strategy concept with the notion of managerial choice, but that is not an accurate representation of the field or of its meaning for organization theory. The central propositions of strategy-policy that

have meaning for organization theory are (1) that selection pressures primarily act on an entity's policy mix, not its organizational structure, (2) that policy changes permit organizations to alter their environments without changing the environment, (3) that the critical contingent aspects of the environment are local rather than global and are functions of the logic underlying the niche structure, and (4) that adaptation is a two-tier affair which demands attention to both niche maintenence and the changing nature of rivalry itself.

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